

Offline pipeline(Data Ingestion Module)

Input -> Unstructured medical documents(PubMed abstracts, review articles, case studies, guidelines, clinical trial reports, pdfs, xml/html docs(optional))

Step1: Parsing and Cleaning(parsing, stripping, section extraction, noise removal, reference removal(optional))

Output -> Structured Text Documents

Step2: Chunking(Fixed-length chunking, semantic chunking, sliding window with overlap)

Output -> Document chunks + metadata

Step3: Embedding(domain specific transformer encoder, embeddings, vector normalization)

Output -> Dense vector representations

Step4: Vector DB and Semantic Indexing(Vector DB(FAISS, Milvus, Pinecone), ANN indexing, metadata storage)

Output -> ANN indexed Vector DB

Step5: Lexical Indexing(BM 25 index, store term frequencies, maintain document IDs)

Output -> BM25 Index

Online pipeline

Input -> User Query

Step1: Query Processing(NLP preprocessing, NER, entity linking, query expansion)

Output -> Processed Query

Step2: Query Embedding(Embedding Model)

Output -> Query Vector

Step3: Hybrid Retrieval(Query vector with vector DB, query text with BM25)

Output -> top k semantic and lexical matches

Step4: Hybrid Fusion + Re-ranking -> Innovation(weighted scoring)

Output -> Final top k evidence chunks

Step5: Context Assembly(Deduplication, **NLI**, evidence level filtering, prompt construction)

Output -> Optimized context block

Step6: RAG(Fine tuned biomedical LLM evidence synthesis)

Output -> Draft output

Step7: Verification & Groundedness(Confidence score, Attribution alignment, detect hallucinations)

Output -> Validated response

Step8: Structured Output(evidence summary, citations, score, insights)

Output -> Evidence based Insights in structured format